



VOICE BRIDGE

Dialect-Aware Voice AI for Rural Health and Farming in Chittagong

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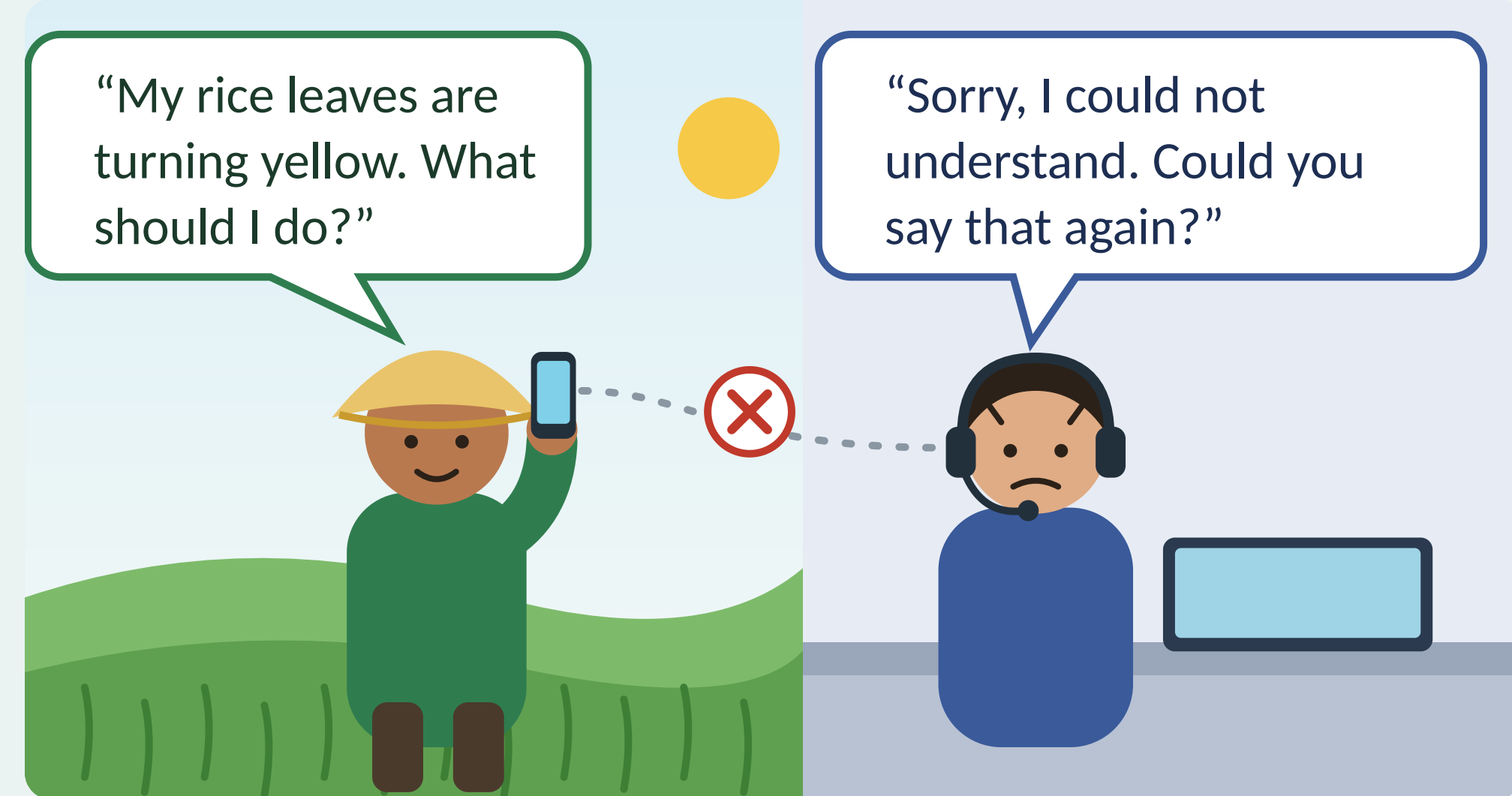
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1. INTRODUCTION

Chittagong is one of the largest divisions of Bangladesh, yet its dialect differs from standard Bangla in pronunciation, vocabulary and rhythm. People from other regions often struggle to follow it, and automated systems struggle even more.

Farmers and patients who phone a call center for crop or medical advice must explain urgent problems in their own speech. If the agent cannot understand them, the call ends without help.



A farmer's call is not understood by the call center. (Illustration)

2. RESEARCH GAP

- Bangla voice services are mostly built and tested on standard speech, so dialects are poorly served.
- Pre-trained speech models are biased toward standard pronunciation and make more errors on dialect speech.
- Earlier work usually targets one part, such as recognition or dialect classification, not a full spoken conversation.

3. OBJECTIVES

- Build a speech recognition model that understands Chittagonian speech.
- Connect it to a language model with an updatable crop and medicine knowledge base.
- Reply in clear, natural spoken Bangla through speech synthesis.
- Offer it over a basic phone call or mobile app, for farmers first and patients next.

7. EXPECTED IMPACT

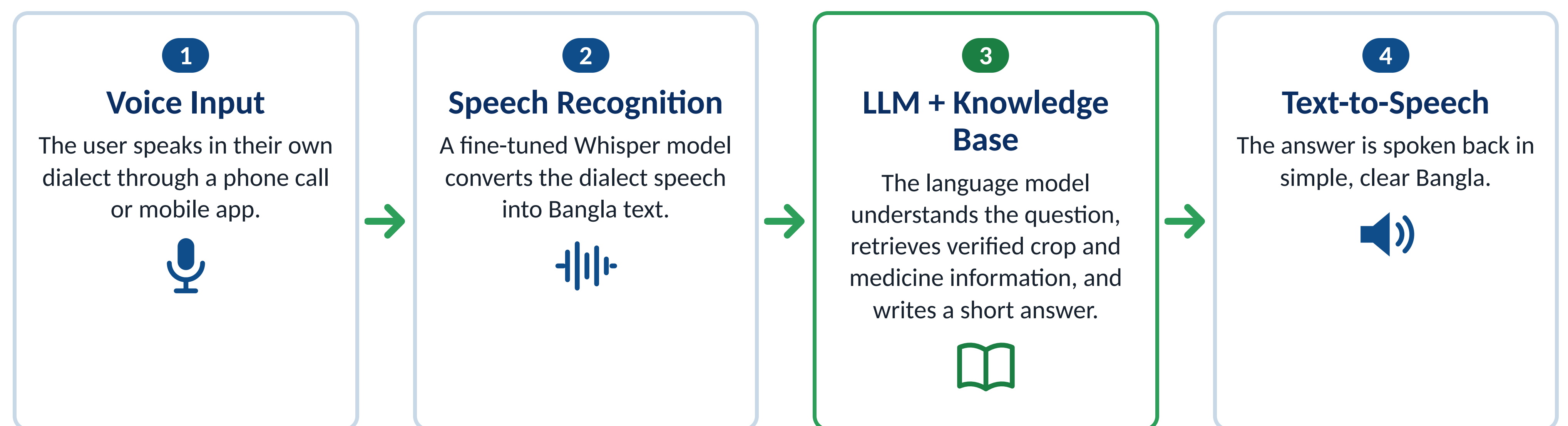
- Farmers get timely advice in their own speech, helping reduce crop loss.
- Rural patients get basic health guidance without traveling.
- Call centers can answer routine questions without dialect-speaking agents.
- Elderly and low-literacy users get help by voice, without changing how they speak.

8. CONCLUSION

Voice Bridge proposes a practical way for Chittagong's farmers and patients to get expert advice by simply speaking. Our early recognition results give a working starting point and show the need for dialect-specific improvement. By combining speech recognition, a language model with a verified knowledge base, and speech synthesis, the system aims to reach people whom today's services often fail to understand.

4. PROPOSED SYSTEM

We propose a voice-to-voice assistant that listens to a question in the Chittagonian dialect and answers aloud, in four stages.



Example: A farmer says the leaves of his rice plants are turning yellow. The system recognizes the question, looks up likely causes and treatments, and tells him aloud what to do. The knowledge base can be updated with new advice without retraining the speech model.

5A. FARMING (PRIMARY FOCUS)

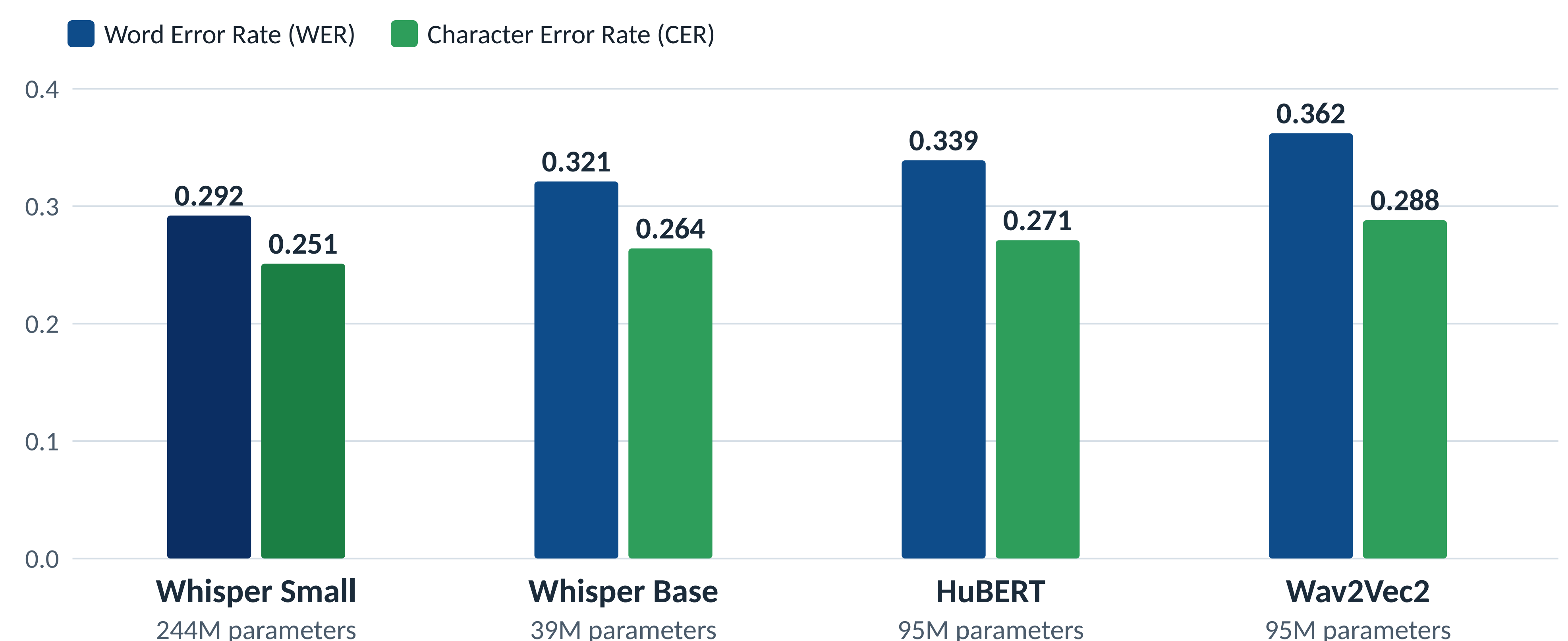
- Identifies crop diseases and pests from the farmer's own description and suggests treatment.
- Gives guidance on fertilizer, irrigation and sowing time.
- Handles routine agricultural call center questions in the caller's dialect.

5B. MOBILE MEDICINE

- Answers common questions on symptoms, medicine use and first aid by voice.
- Advises when to see a doctor; it does not replace one.
- Reaches people far from clinics on a basic phone.

6. STARTING POINT: PRELIMINARY ASR RESULTS

As a foundation for stage 2, we fine-tuned four pre-trained speech models on regional Bangla speech. Lower error is better.



0.292

Lowest WER, achieved by Whisper Small, so it is our starting model for stage 2

~60%

Accuracy of our baseline model trained only on Chittagong speech, limited by small training data

CER is lower than WER for every model: the models capture most word forms but miss regional sound patterns.

9. FUTURE WORK

- Collect more Chittagonian speech to improve recognition accuracy.
- Build the knowledge base with agriculture and health experts, then integrate the language model and speech synthesis into a prototype.
- Test with farmers, patients and call center staff, and extend to other regional dialects.